Nicolas Corfmat

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EDUCATION

University of California, Santa Cruz — **Santa Cruz, CA**
Bachelor of Science in Computer Science: Computer Game Design — *Sept 2021 - June 2025*
Cumulative GPA: 3.86/4.00
Dean's Honors List: 2021-2025

Relevant Coursework: Computational Models, Data Structures & Algorithms, Web Applications, Game Technologies

SKILLS

Software/Programming: C++, C, C#, Python, Java, JavaScript, TypeScript, HTML, CSS, React, Vue, Py4Web, Git, Unix
Engines & Tools: Unity, Unreal Engine, Phaser, Blender, Adobe Premiere Pro
Languages: Fluent English, Fluent Spanish, Conversational French

PROJECTS

Shape Up - Senior Capstone Project — April 2025 - June 2025
University of California, Santa Cruz
- Shipped a mobile iOS/Android puzzle game in Unity and C# with a 5-person team, implementing levels and UI.
- Integrated Analytics SDK to track player and device metrics for analysis, syncing data with Unity Cloud services.
- Profiled builds and optimized performance using object pooling, simple colliders, singletons, and asset compression.
- Led iOS release pipeline, securing next-day App Store approval, and managed internal/public builds via TestFlight.

Recipe Sharing App - Full-Stack Developer — May 2025 - June 2025
University of California, Santa Cruz
- Built a full-stack recipe website with Py4Web (Python) and Vue.js, using HTML/CSS for a responsive UI.
- Designed and implemented RESTful APIs to perform CRUD operations on recipe and ingredient data.
- Connected frontend with backend APIs, enabling multi-field recipe search by name, type, and ingredients.
- Enforced author-only recipe editing with server-side auth to prevent unauthorized changes to other users' content.

Real-Time WebGL Demo - Solo Developer — *Feb 2025 - March 2025*
University of California, Santa Cruz
- Developed an interactive 3D demo using WebGL and GLSL shaders for diffuse, specular, and spotlight lighting.
- Created dynamic lighting controls (position, color, enable/disable) using HTML and CSS with live scene updates.
- Built a camera system with full 3D navigation (forward/back, strafe, pan, tilt) using vector math and rotation matrices.
- Optimized texture loading, UV mapping, and materials through efficient GLSL uniform and buffer handling.

SpaceMine - Lead Programmer — *Nov 2024 - Dec 2024*
University of California, Santa Cruz
- Spearheaded development of a 2D mining simulator using TypeScript and Phaser, with Vite for rapid deployment.
- Serialized game state into a single contiguous byte array, allowing for efficient save/load and undo/redo operations.
- Incorporated modding capabilities via external DSL and offered game as installable offline desktop and mobile PWA.
- Localized the game to three languages, dynamically loading in-game text from a JSON file at runtime.

Chess vs. AI - Solo Developer — *Nov 2023 - Dec 2023*
University of California, Santa Cruz
- Built a Chess AI in C++ to compare thousands of possible moves and strategically execute the optimal move.
- Utilized minimax algorithm, alpha-beta pruning, and transposition tables to significantly reduce AI search complexity.
- Developed an intuitive graphical interface using the IMGUI library, enhancing the user experience.
- Incorporated the Universal Chess Interface (UCI) to enable the AI to play against other chess engines.